

Important Notes

1) Hessian is second derivative of loss function
distance $\rightarrow$ Speed $\rightarrow$ acceleration (Analogy)

Regression $\text{loss} = (\hat{y} - y)^2$
 $\frac{\partial^2 \text{loss}}{\partial y} = 2$ or 1 depending on imp
gradient $= (\hat{y} - y)$

Classification $= -(y \log(p) + (1-y) \log(1-p))$
 $g = p - y$
 $h = p(1-p)$

2) Impact of lambda (λ) (regularization)
as per weigh formula

$$w = \frac{G}{K + \lambda}$$

large lambda small correction

How aggressively can leave value change prediction

3) Impact of γ
in gain calculation
 γ prevent further split

High γ

Fewer split

Shallower tree

Simpler model

less overfitting